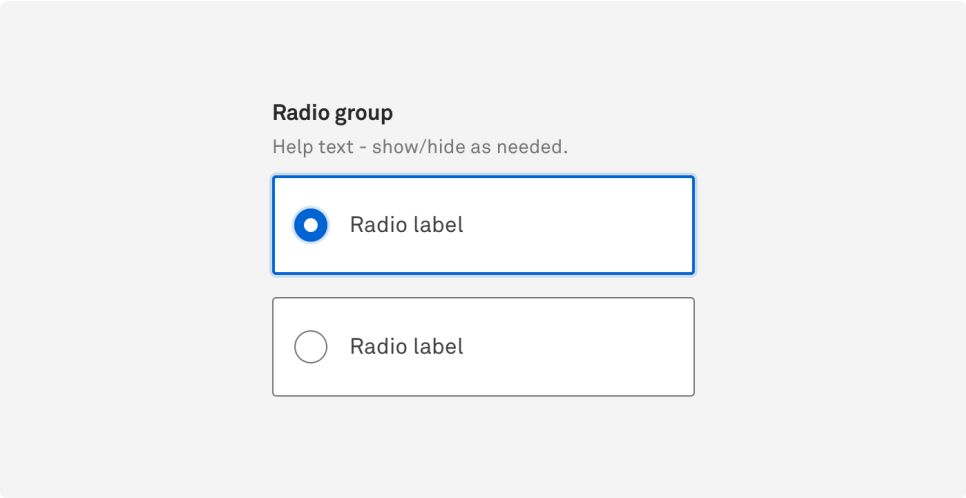
*Need help? Self-Service || Yammer || **No new spaces from Nov-23rd***

Radio Group

Created by Tony Wang, last modified on 10 Mar 2021



A radio group allows users to select exactly 1 option from the small set of options.

Version

2

Availability

Web

On This Page

- Usage Guidelines
- Improved UI
- Anatomy
- Variants
- Supporting Your Selection Optior
- Offer Smart Default
- Sorting Your Options
- States
- Radio Button Interactions
- Best Practices
- Content Guidelines
- Accessibility Considerations
- Interactions
- Capabilities
- Related Patterns

Useful Docs

- Offering Smart Default Option
- When to Use Compact or Comfortable
- Using Images to Support Text Labels
- Checkbox & Radio UI Improvement
- Interaction for Selection Controls

Specs

- Radio Button (Web)
- Form Labels (Web)

UI Playground

- Web Playground
- React Playground

Usage Guidelines

When to use Radio Groups:

- When you want user to make **exactly 1 selection** from a list of options.
- When **seeing ALL available options** is critical.
- When the options are mutually exclusive.
- When there is a minimum of 2 options for users to choose from.

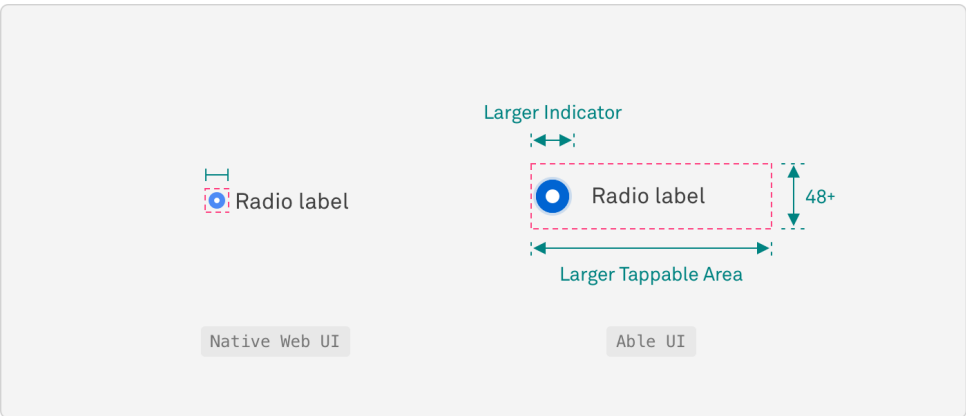
Recommendation (Soft Rule):

- Not to have too many options in Radio Group, ideally a max of 5 options is acceptable, consider Select for >5 options.

Don't use a Radio Group if:

- You want user to select more than 1 option. Use Checkbox Group instead.
- You have more than 5 options, consider using Select.
- The user needs to choose between Yes/No of a setting, generally a standalone Checkbox is more suitable.

Improved UI



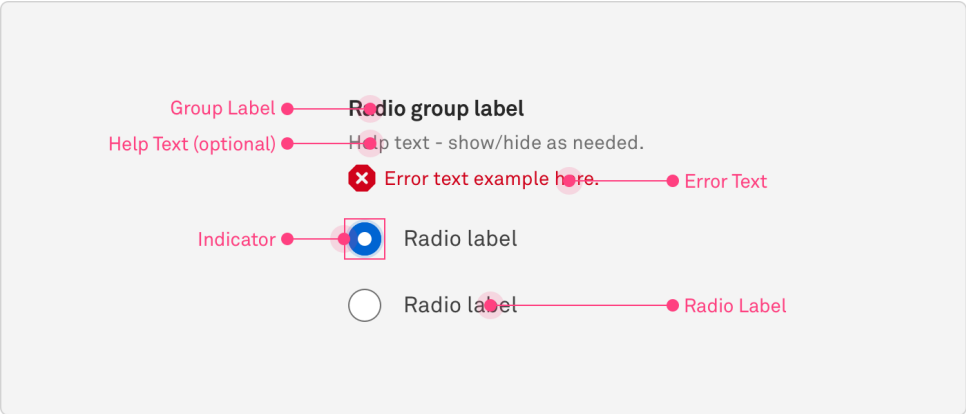
With native radio controls, the indicator is small and the tappable area is limited to just the icon visual.

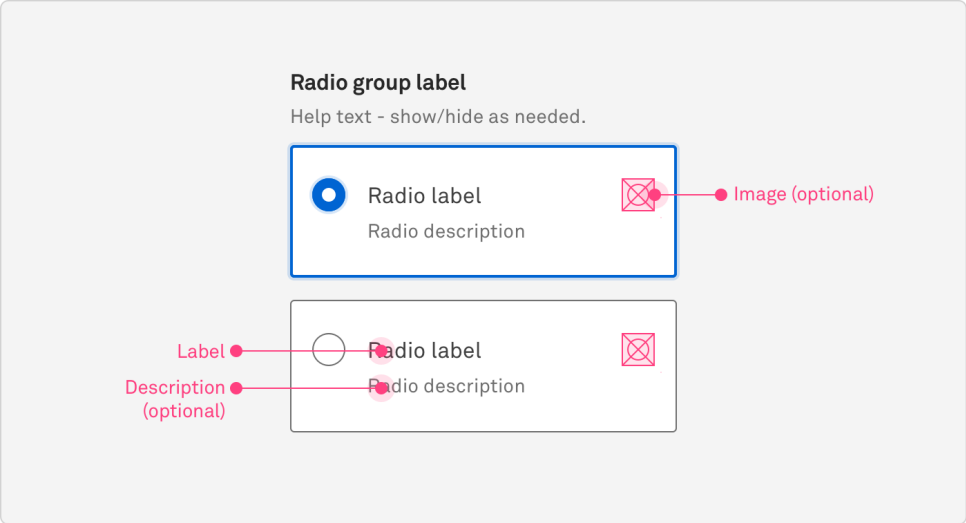
In order to improve the user interaction of our radio button, we have:

- Increased the indicator size more than 200%
- Increased the tappable area to allow selection by tapping on the label (for touchscreen devices or mobility impaired users)

For more details behind this UI improvement, see Improving Checkbox & Radio Indicators.

Anatomy



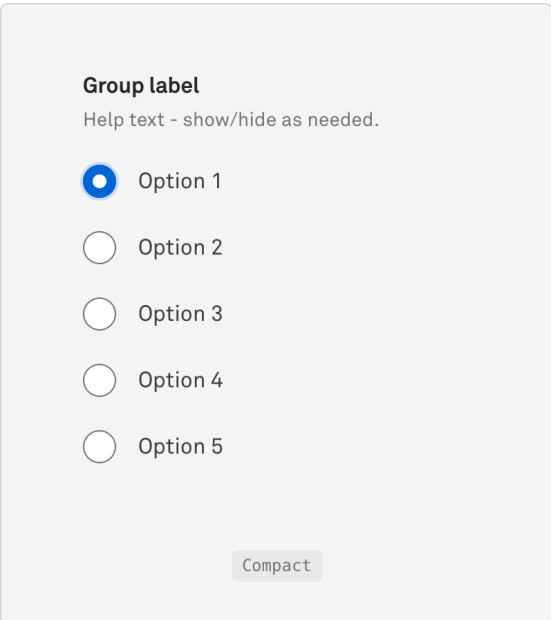


Variants

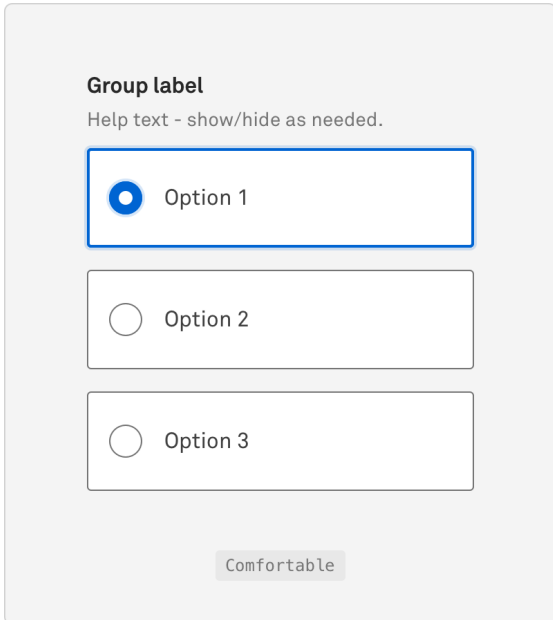
There are 2 variants to choose from — *Compact Group* and *Comfortable Group*.

For guidance on the which variant to use, see When to Use Compact or Comfortable.

Compact Group



Comfortable Group



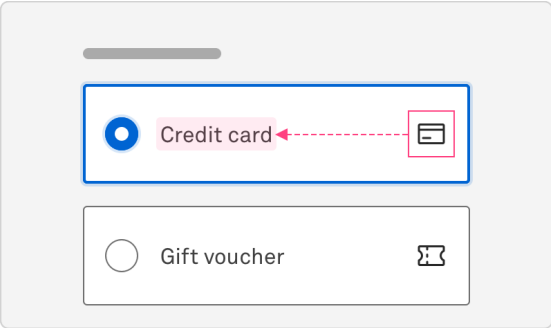
Designed to help user scan through ALL the available options and **make a simple decision quickly**.

Designed to guide user to **make an important (or a complex) decision** by putting more emphasis on the option user is selecting.

Supporting Your Selection Options

In the Comfortable Group variant, you have the ability to:

Add a Supporting Image



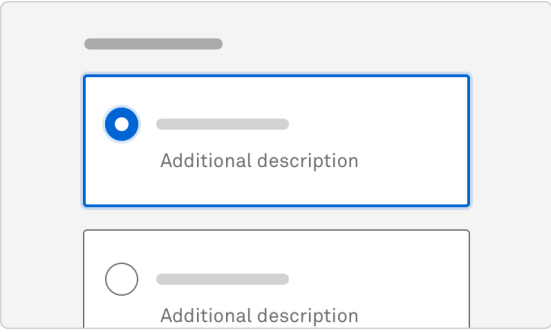
Adding a supporting Image can **help convey the radio option quickly to our users**.

To learn more about choosing a supporting image, see Using Images to Support Text Labels.

Image types that can be used in Radio Group:

- Icon in 24x24
- Pictogram in 56x56

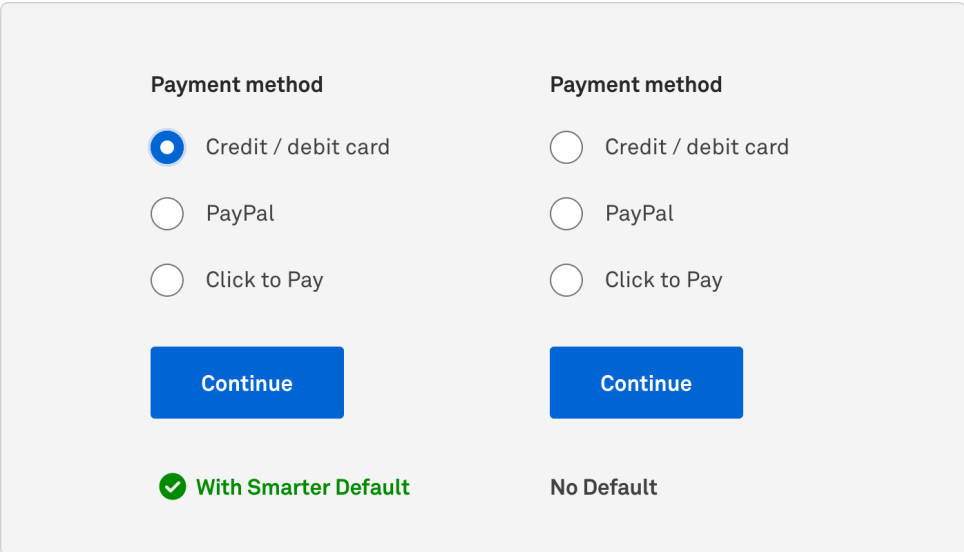
Add an Additional Description



Additional description provides **more clarification on the options** you're providing.

For more guidance on additional description, see Checkbox Description in Content Guidelines.

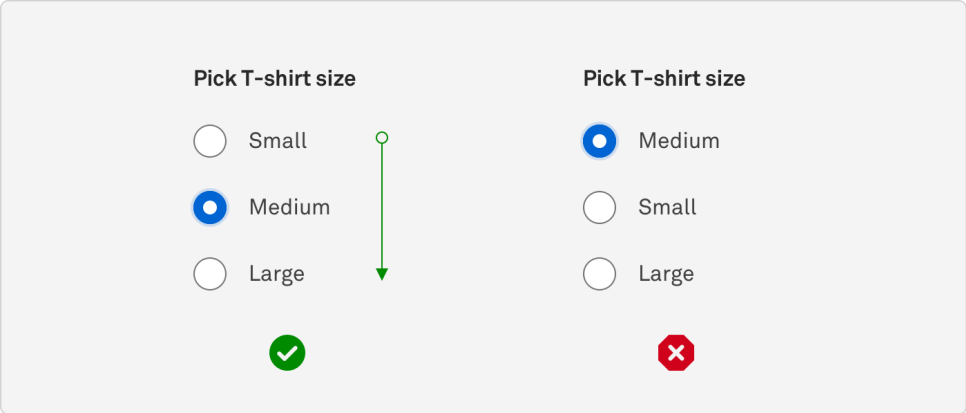
Offer Smart Default



A *Smart Default* is an option that is selected by default, and is the most desired or frequently selected option based on user's context and needs.

See Offering Smart Default Option for more detailed guidance.

Sorting Your Options



When we present a list of radio options, the order should be familiar to the users at first glance.

Presenting an unfamiliar order will **increase user's cognitive load** and they could **potentially make an incorrect selection**.

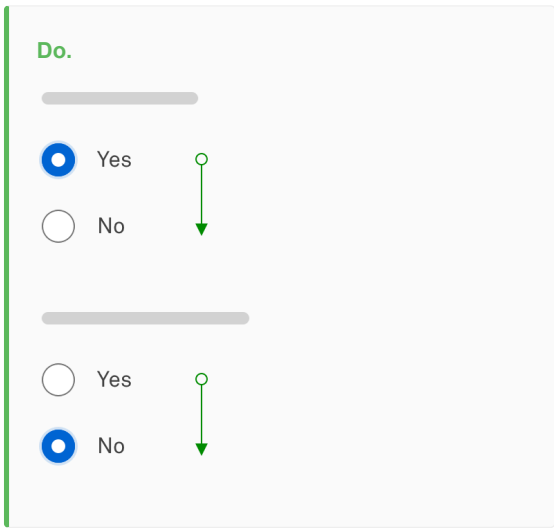
Therefore, designers should design the options to be presented in an order a user expects and can make sense of.

Ordering Rules

- Ascending or descending
- Most common ↔ Least common (vice versa)
- Smallest ↔ Largest (vice versa)
- Safest option ↔ Riskest option

Keep in mind

Be consistent when listing the same type of radio options within the same page.



Even though the default selection changes, users can expect 'Yes' to always be the first option on a form or webpage and make their choices based on this expectation.

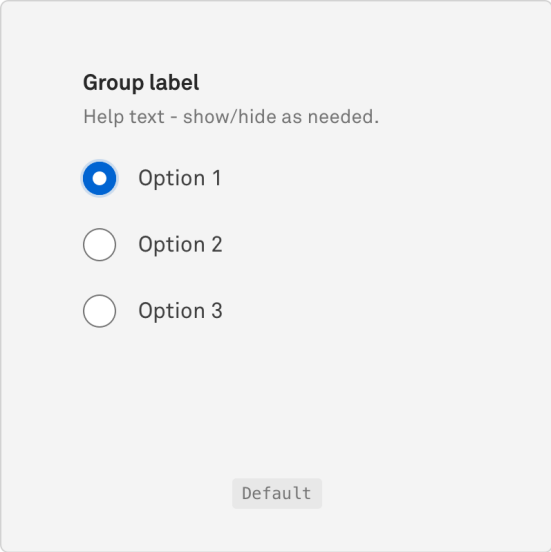
The default selection can constantly switch between 'Yes' and 'No', which is an inconsistent experience on a form or webpage.

States

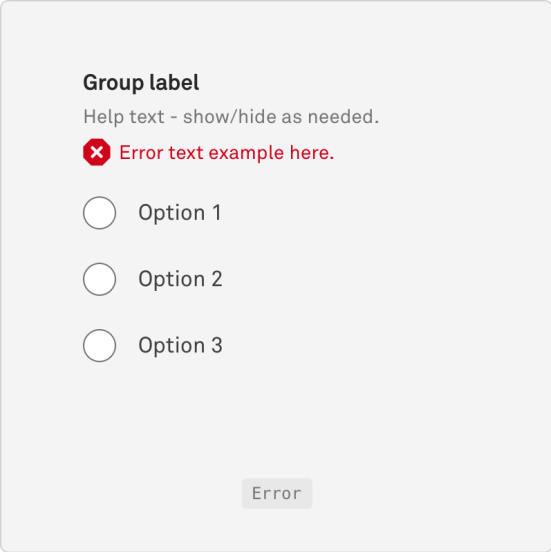
Each Radio Group has a Default and an Error state.

Default State

Error State



Users can make their selection freely in the Default state without any interruption.



Error state will only appear when user submits the form and non of the option are selected.

The error should disappear immediately when use has made a selection.

Radio Button Interactions

There are 8 unique interaction states for each Radio Button.

A radio button can be:

- Unselected
- Selected

Each radio button has 4 different states

1. Normal
2. Hover
3. Pressed
4. Focused

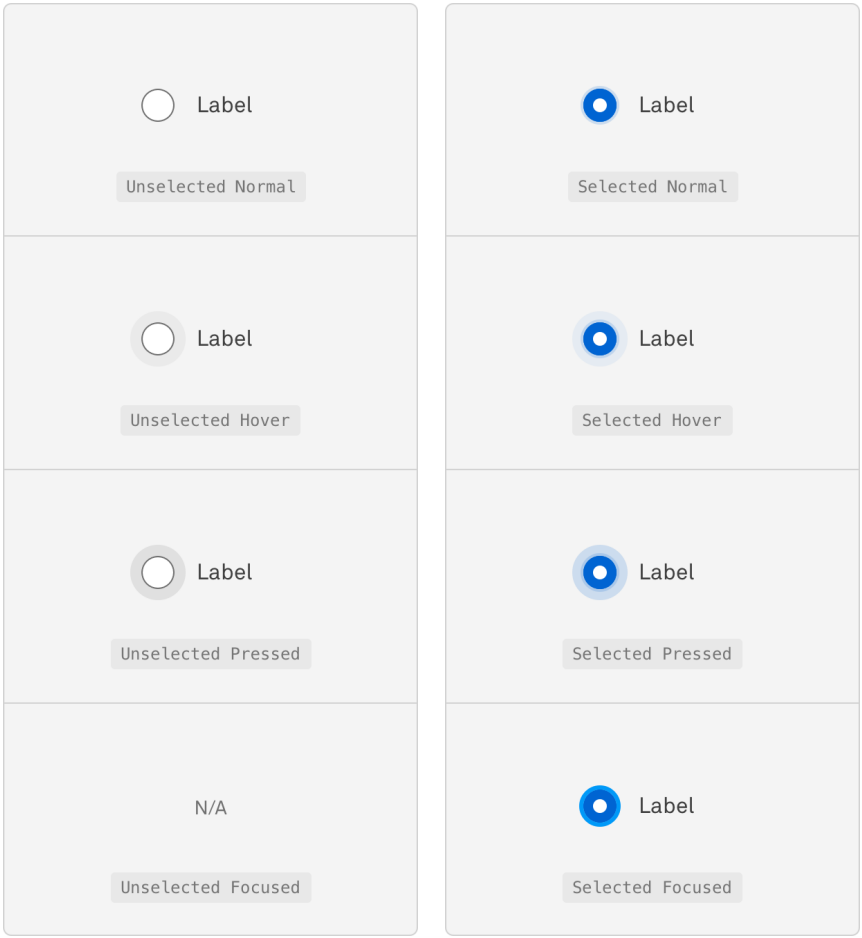
Keep in mind:

- '*Disabled state*' is not supported in radio button. See why we don't support disabled state.

To find out more details on the different states, see Interaction States.

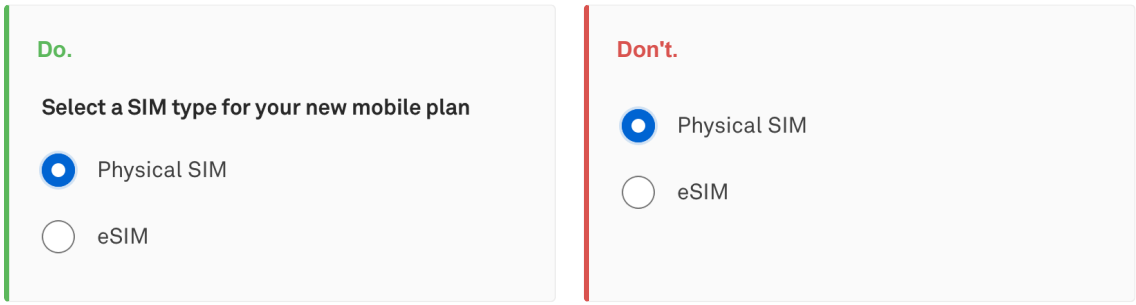
Unselected States

Selected States

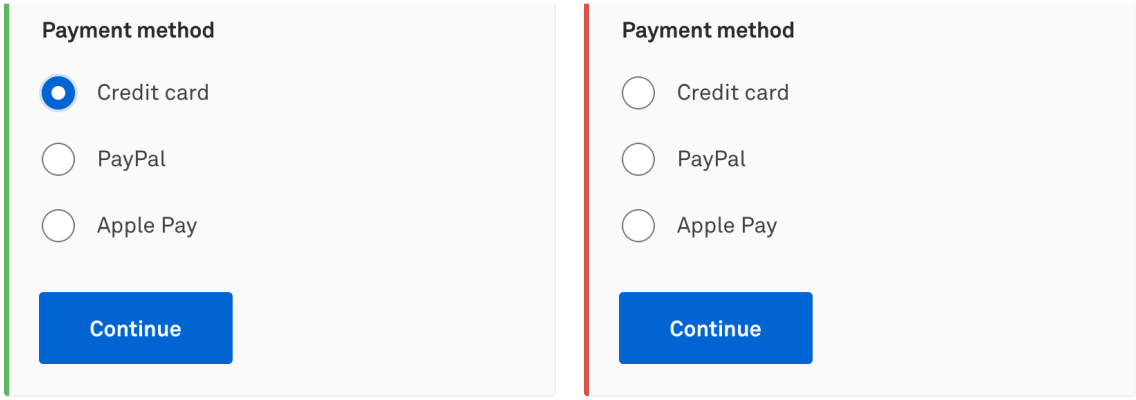


Best Practices

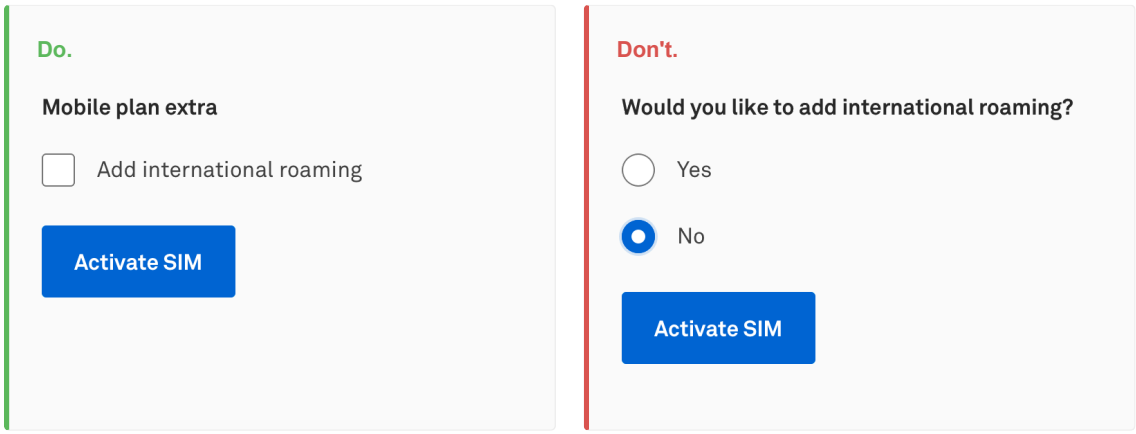
- **Each Radio Group must have a Group Label.**
  Group label should clearly describe what the list of options represent. This is important for accessibility, since a screen reader will read out the group label with each radio option. Don't assume that the radio options are self-explanatory. See Content Guidelines for copy guidance.

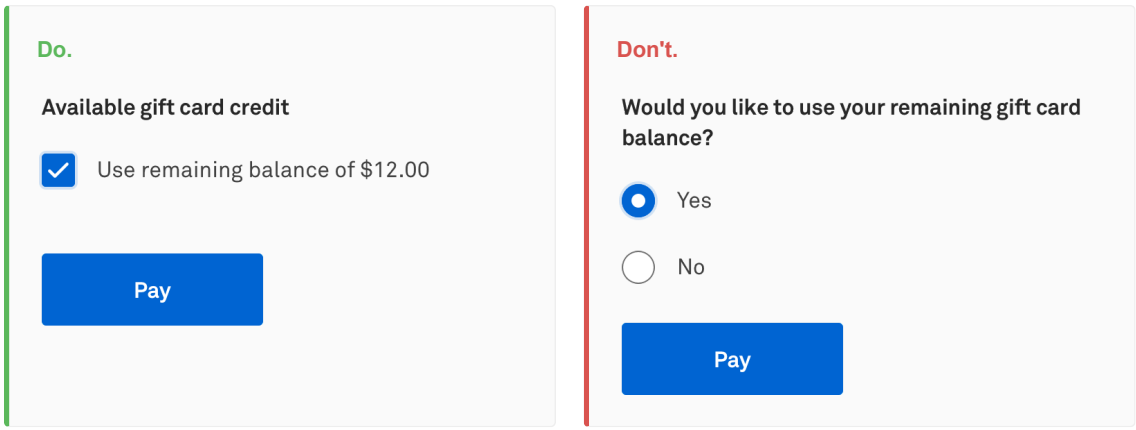


- **Spend time on research. Pick the most appropriate default selection for your users.**
  Make use of analytics and historical data. If you know majority of the users are likely to select a credit card option in your feature, it's better to make that option the default selection to help user go through their task faster.

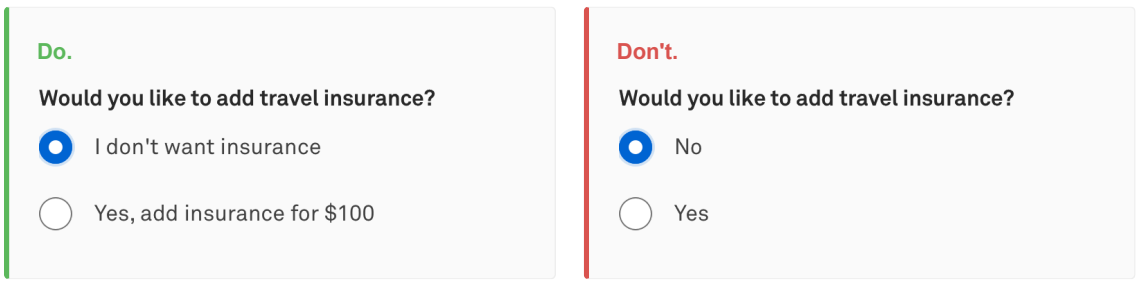


- **Consider using a standalone Checkbox if we know which option user is likely to pick in a Yes/No question.**
  Using standalone checkbox as Opt in/out options instead of Yes/No question provides better sentiment than explicit Yes/No radio options. See examples below:

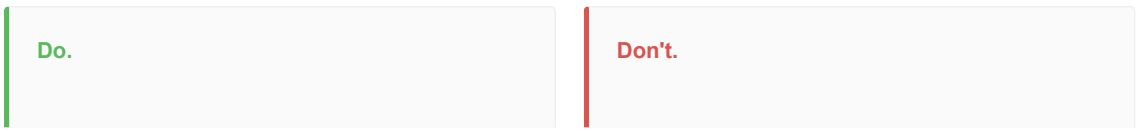


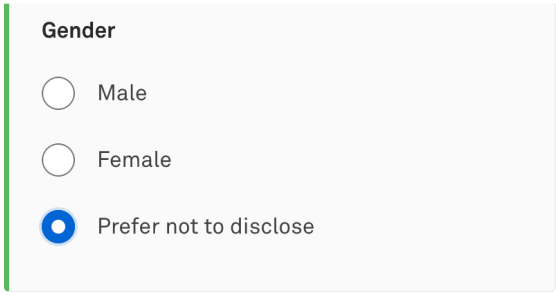


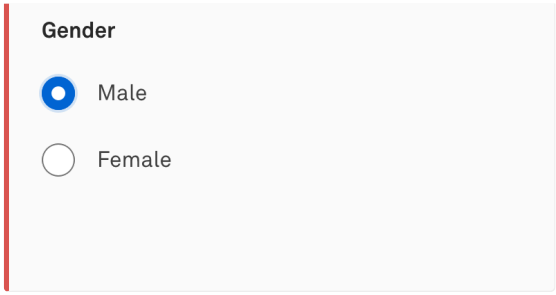
- **Avoid only writing 'Yes/No' as radio options, be more descriptive.**
  This follows the same principle where we don't recommend using 'Yes/No' in Action Button. 'Yes/No' labels typically don't provide clear indication on what will actually happen when users select a specific option. Users often need to memorise/read the question again, which may lead to interpreting the option incorrectly. Therefore, we **recommend to be descriptive when writing the options**. See Content Guidelines for more guidance.



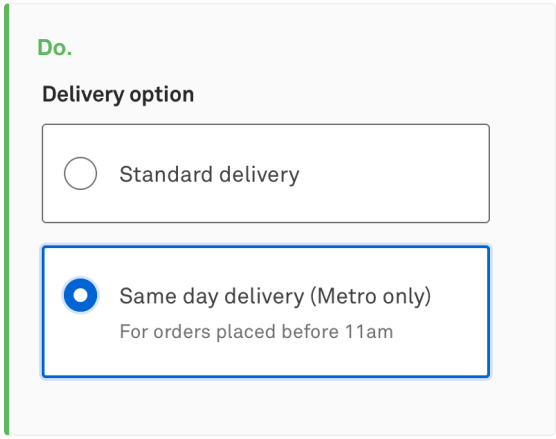
- **Don't use radio button to initiate actions.**
  Radio group presents a list of options, they're not action buttons. Once the user has made the selection, they must click a submit/save button to make the desired changes.
- **Be inclusive with the list of options.**
  If you're asking a user what their gender is, don't offer binary options only. Be sure to provide other possible options to be inclusive.

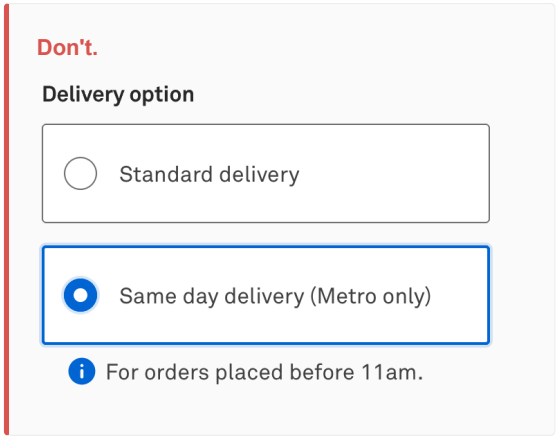




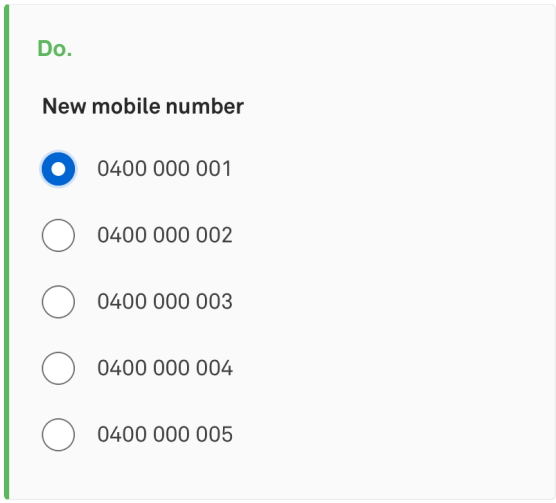


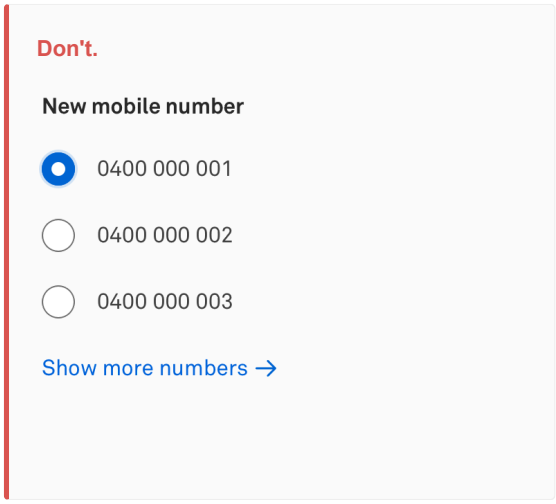
- **Don't hide information under a radio button.**
  All info relevant info should be **presented up front** to allow user to make a selection. Don't surprise them with critical information after their decision.



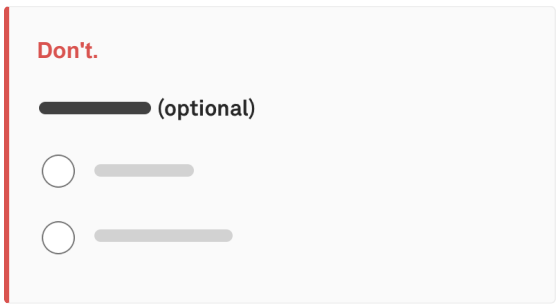


- **All options should be visible at all times.**
  One of the characteristics of Radio Group is to display ALL options to users to make a selection. Don't hide options.

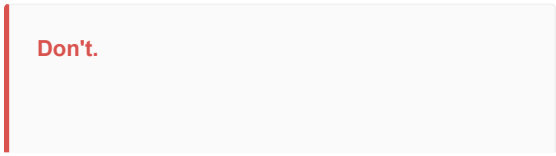


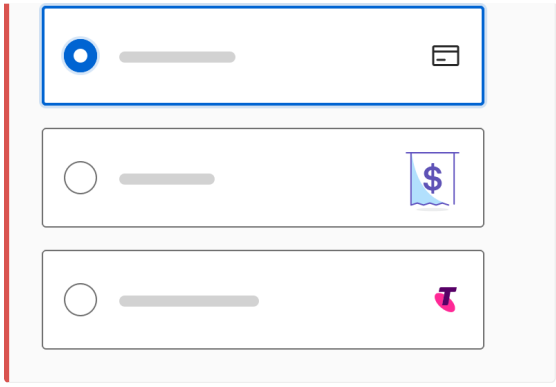


- **Don't mark radio groups as optional. They are always required.**
  Radio group should always be a required selection with exactly 1 option to be selected.

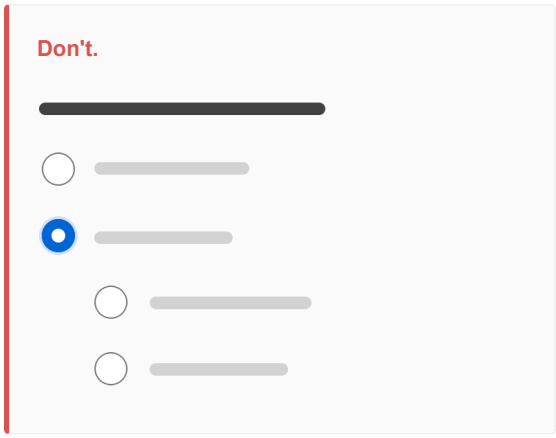


- **Don't combine different image types in the same Radio Group.**
  The decorative images should be the same type within the same group.

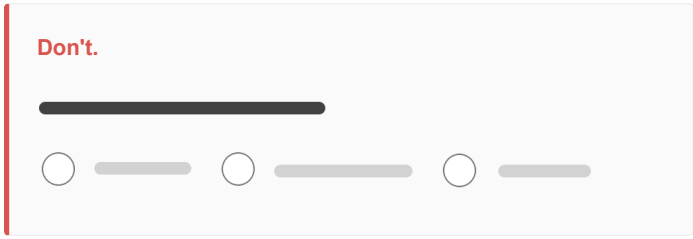




- **Don't nest radio buttons.**



- **Don't stack radio buttons horizontally, always stack vertically.**
  Compact & Comfortable variants should always be stacked vertically.



Content Guidelines

This section provides guidance on how to write effective copy for each text elements (also documented in Telstra's Content Style Guide):

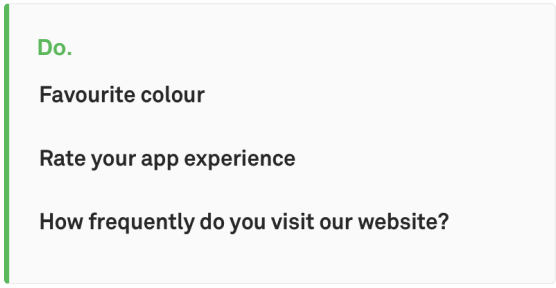
1. Group Label
2. Help Text
3. Error Text
4. Radio Label
5. Radio Description

1. Group Label

A radio group **always** contains a group label which clearly describes what the list of options represent. Don't assume the radio options are self-explanatory without a group label.

When writing a group label:

- Consider the most appropriate and consistent way to phrase it - as a question, statement, phrase or single word.
- Clearly describe the grouping category for context and clarity. Keep it succinct, no longer than a single sentence/question.



- Use singular form, not plural, to indicate that user should select only 1 option.

Do.	Don't.
Shipping method	Shipping methods

- Don't use any punctuation (unless the group label is phrased as a question).

Do.	Don't.
Which is your favourite colour?	Which is your favourite colour:

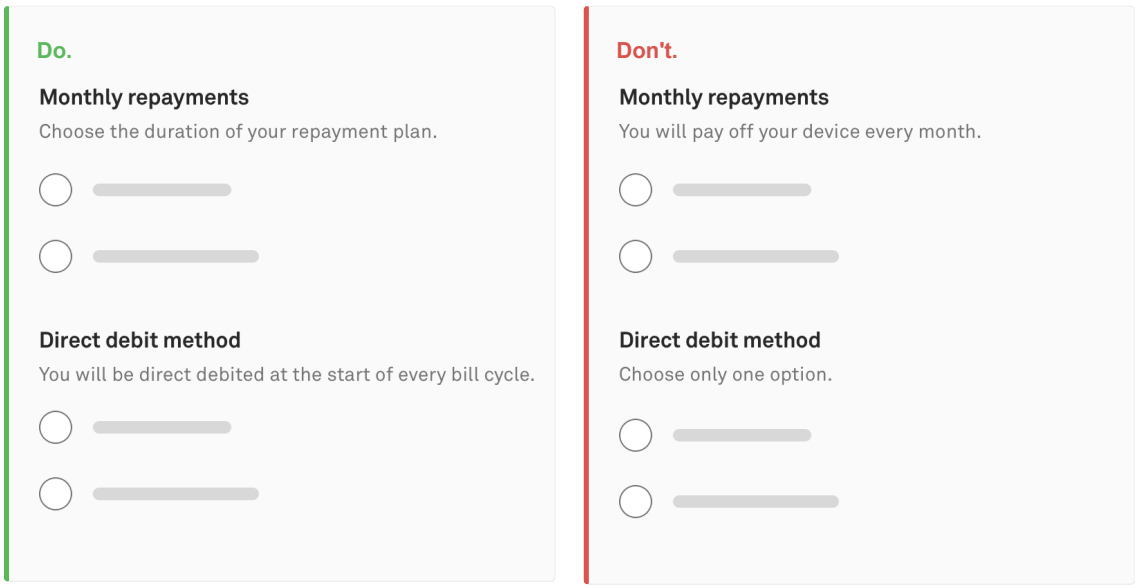
- Keep it consistent with the other labels in Text Field and Checkbox Group within the same page.
- Use sentence case.

2. Help Text

Help text is the optional text displayed below the group label. It should only be used to provide assistance to the user if they require more context or instruction to aid them with their selection.

Help text should:

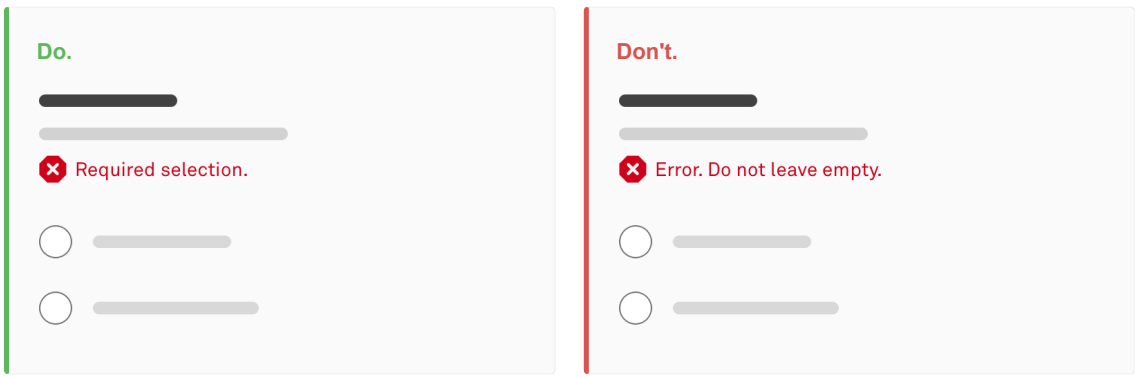
- Only be used if it's necessary to provide specific instruction or context to users.
- Be kept short and concise, keeping it to 1 line or sentence of text where possible.
- Be written as a sentence with proper punctuation.



3. Error Text

In a Radio Group, an error will be only be shown if they fail to make a selection and no default selection was provided.

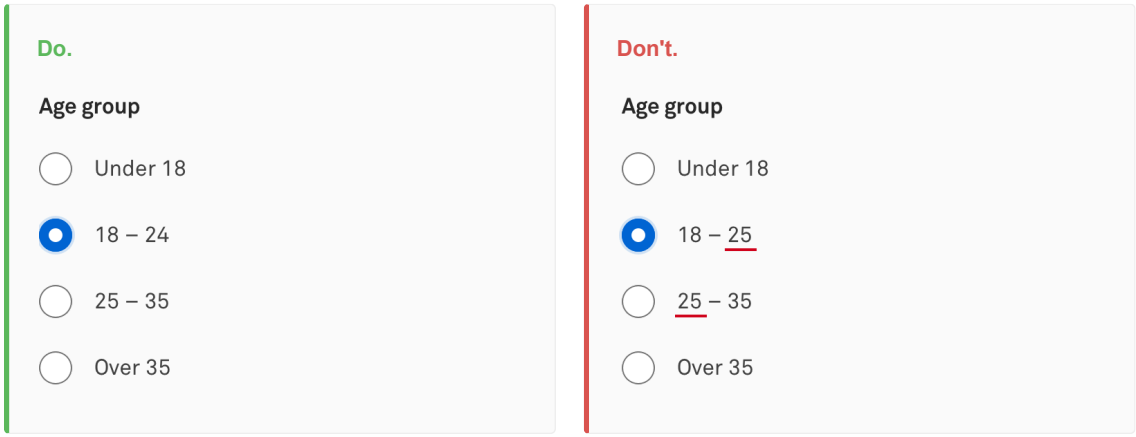
Error text should always be '**Required selection.**'.



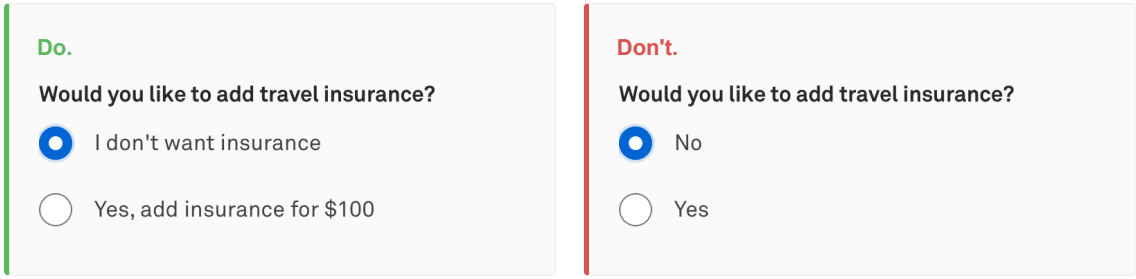
4. Radio Label

Each radio button must have a Radio Label, which should make sense as individual items and as part of a whole radio group. They should:

- Indicate mutual exclusivity from other labels. Be clear to users to select 1 of the distinct options. Don't overlap as in the example below.



- Provide added description for 'Yes/No' decisions to make the outcome of the selection clearer to the user.



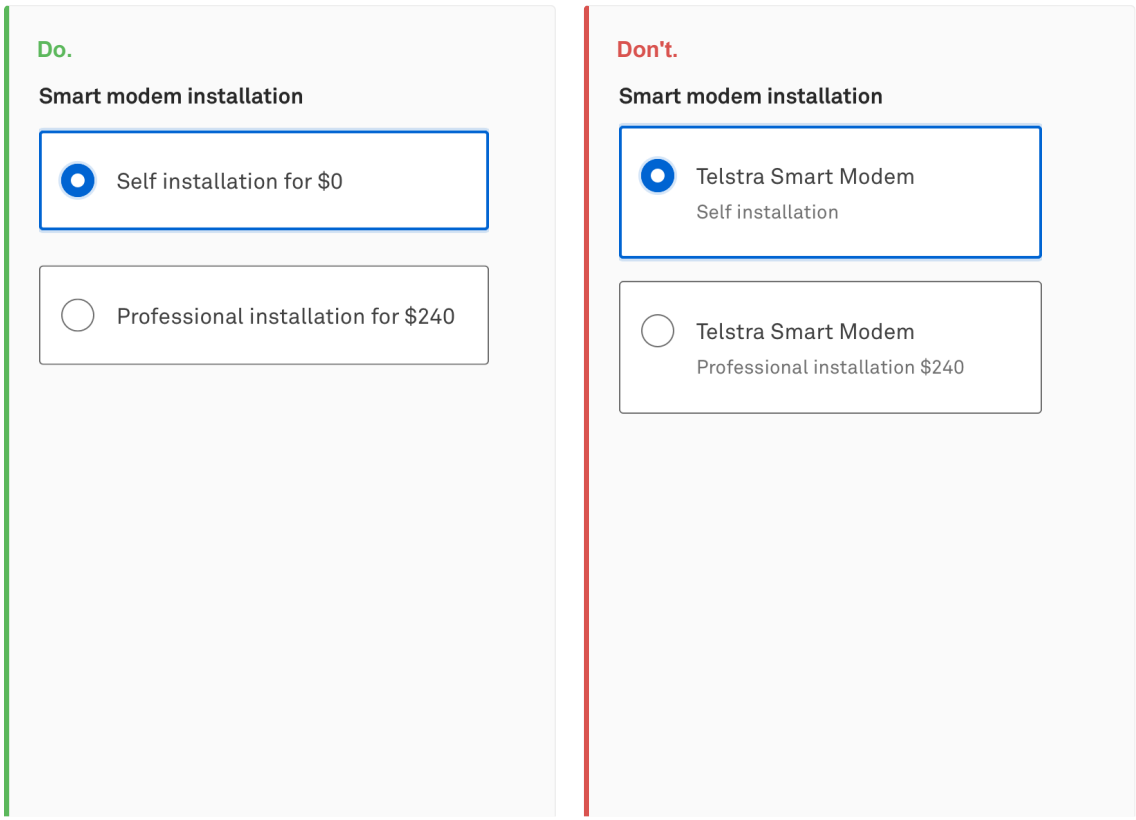
- Be consistently phrased in a list. Don't mix single words and sentences.
- Use the same case on every radio label (e.g. sentence case, title case).
- Avoid using punctuation. If punctuation is required, maintain consistency across all radio labels.

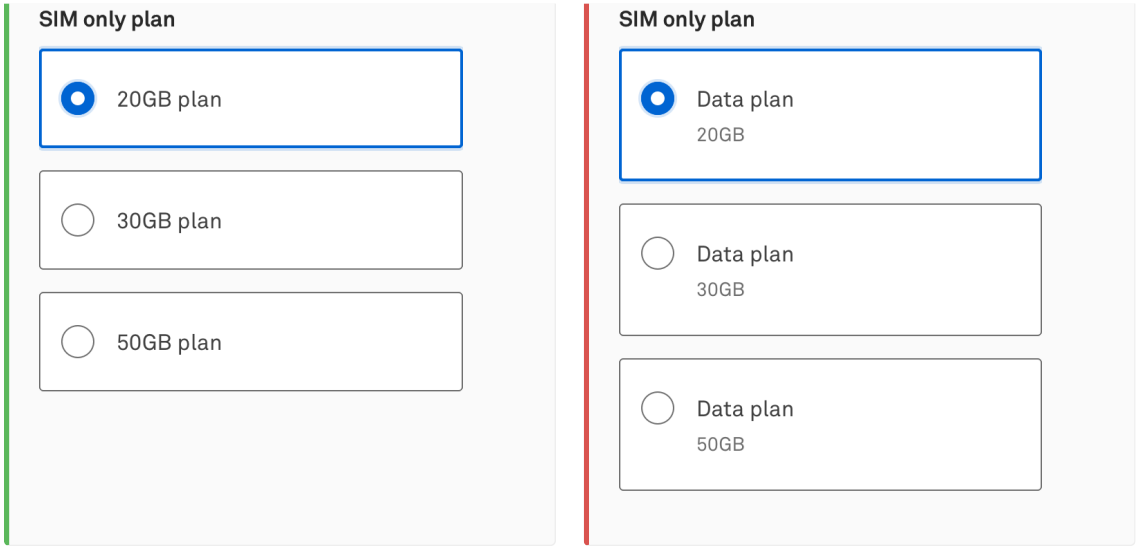
5. Radio Description

Radio Description is optional to support a Radio Label if it will aid the user with their selection. Carefully consider whether additional description will add value to users or whether the Radio Label is sufficient on its own.

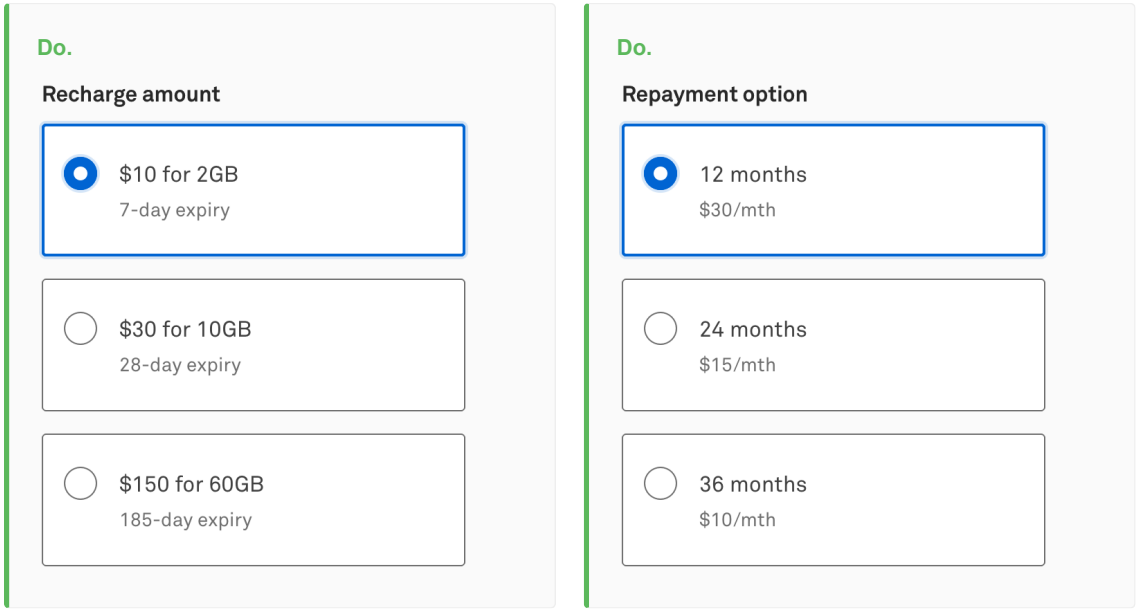
> ⚠ For accessibility, screen readers will read out the Radio Label and Radio Description together. Both should be kept succinct to avoid cognitive overload for the user.

- Critical information should be in the Radio Label instead of the Radio Description, especially if it helps users distinguish between the different options within a radio group.

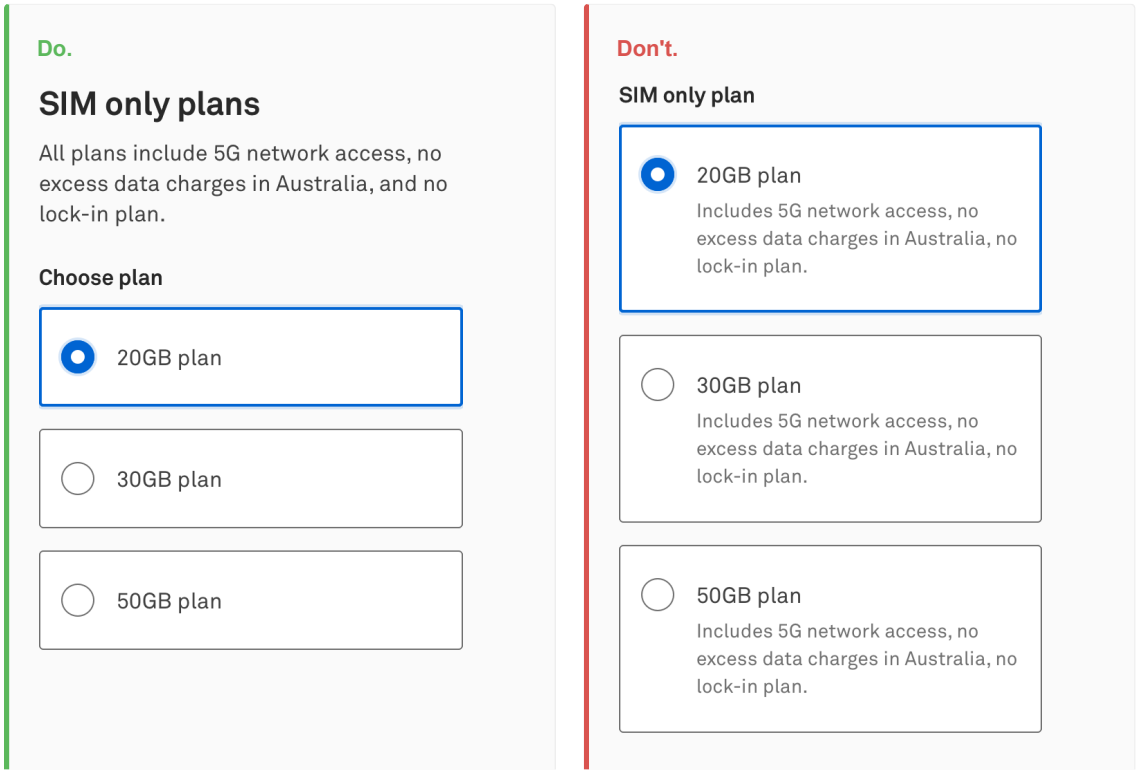




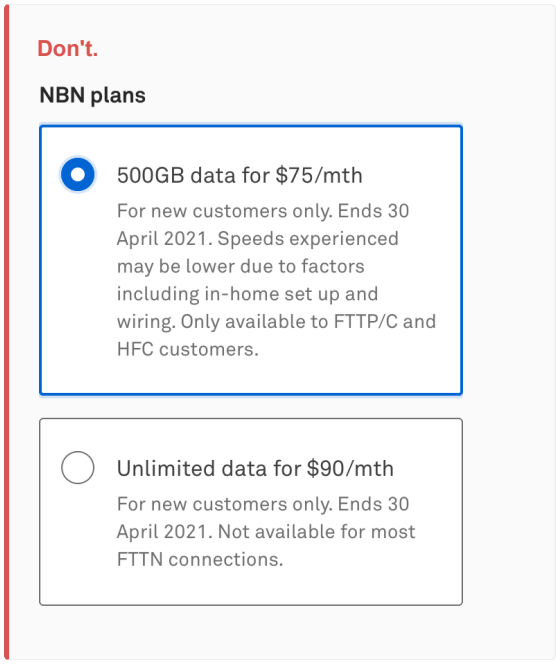
- Only provide additional description if it will help users make a decision (e.g. by adding important conditions to a product or service).



- Don't duplicate the same description for all radio buttons. If there is information relevant to the entire radio group, consider adding this before the group.



- Avoid adding excessive description within each radio button (e.g. fine print or product T&Cs).



- Always use sentence case.
- Include proper punctuation when writing in full sentences.

Please refer to Telstra's Content Style Guide.

Accessibility Considerations

Radio Group is designed to meet the WCAG 2.1 Success Criteria outlined below. All requirements should be considered and understood when using the pattern.

> ⚠️ indicates accessibility requirements will need to be met at time of usage, such as when authoring.

- ⚠️ 1.1.1 Non-text Content (A) - Radio group icons are decorative and will be ignored by assistive technology
- 1.3.1 Info and Relationships (A) - Radio buttons contain a label that is programmatically associated with the field and use semantic grouping.
- 1.4.3 Contrast Minimum (AA) - All text meets the 4.5:1 contrast ratio
- 1.4.4 Resize Text (AA) - When resized to 200%, text is not truncated
- 1.4.11 Non Text Contrast (AA) - Aspects of the radio button required for identification meet the contrast ratio of 3:1
- 1.4.12 Text Spacing (AA) - Text spacing can be increased without loss of content
- 2.1.1 Keyboard (A) - Radio buttons are keyboard accessible
- 2.4.3 Focus Order (A) - Radio buttons follow a sequential focus order
- ⚠️ 2.4.6 Headings and Labels (AA) Group label and radio text have descriptive labels
- 2.4.7 Focus Visible (AA) - Radio buttons have a clear visible focus state
- 2.5.3 Label in Name (A) - The accessible name of the radio button and group label matches the visible label
- 2.5.5 Target Size (AAA) - The touch target size is at least 48 x 48 (dp/pt/px)
- ⚠️ 3.3.1 Error Identification (A) - Radio buttons that are mandatory can be determined in text and in code
- ⚠️ 3.3.2 Labels or Instructions (A) - Labels and instructional text are clear and unambiguous
- ⚠️ 3.3.3 Error Suggestion (AA) - Clear and unambiguous information is provided to help users correct input errors when they occur
- ⚠️ 3.4.2 Consistent Identification (AA) - Radio buttons that capture the same data also contain the same label
- 4.1.1 Parsing (A) - Radio buttons use valid HTML/markup/code
- 4.1.2 Name, Role, Value (A) - Group label, radio text, instructions and errors are marked up correctly to expose the correct name and role to assistive technology

Keyboard

Interaction	Keystrokes
Navigating / toggling radio buttons	↑ / ↓ or ← / → - select an option. TAB - move to the next interactive element outside of the Radio Group.

- Users must be able to navigate to and interact with a radio button using keyboard alone.
- A clear visible indicator should also be present when the radio button has focus.

Screen reader

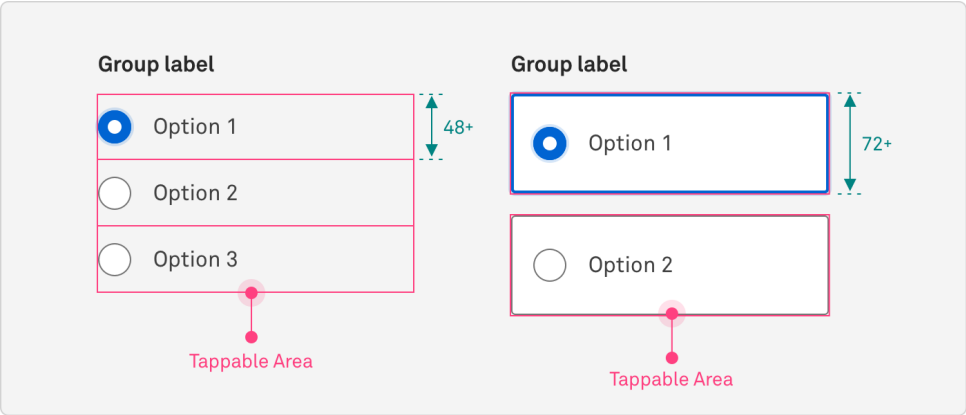
Screen reader behaviour and the ordering of announcements will vary depending on the platform/browser/screen reader combinations being used.

On screen reader focus or toggling the state of radio button, announcements must include:

- The **group label** (<legend>) Depending on the configuration, some screen readers read out the legend either *with every form element*, *once*, or, rarely, *not at all*.
- The **label** of the radio button
- The **role** ie "radio button"

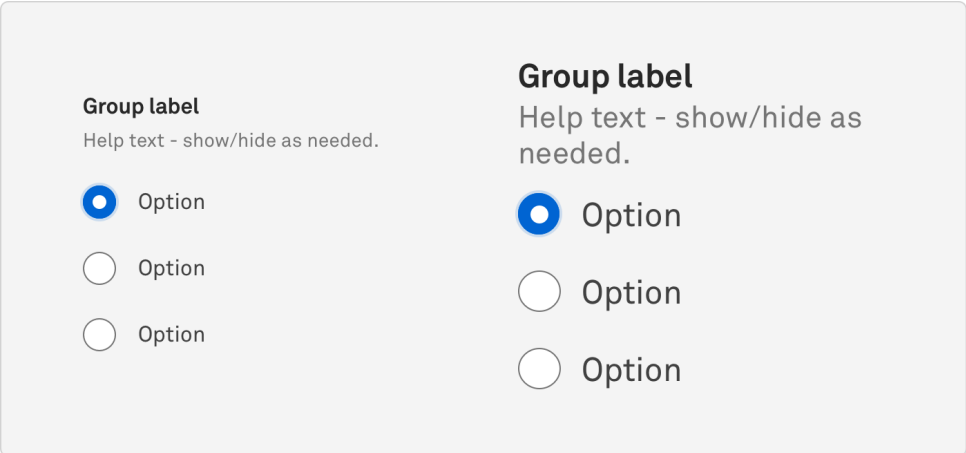
- The **state** of the radio button if selected, ie "Selected" Depending on screen reader, some will also announce "Unselected"
- The **ordering** of radio button within the group, ie "1 of 3"

Interactions



You can use designated key strokes to select an option and navigate between the options.

Capabilities



Robust Layout.

Components are scaled automatically based on the font size and zoom level in user's browser settings.

Related Patterns

- Checkbox
- Checkbox Group
- Select
- Multi Select Row
- Single Select Row

No labels

Atlassian